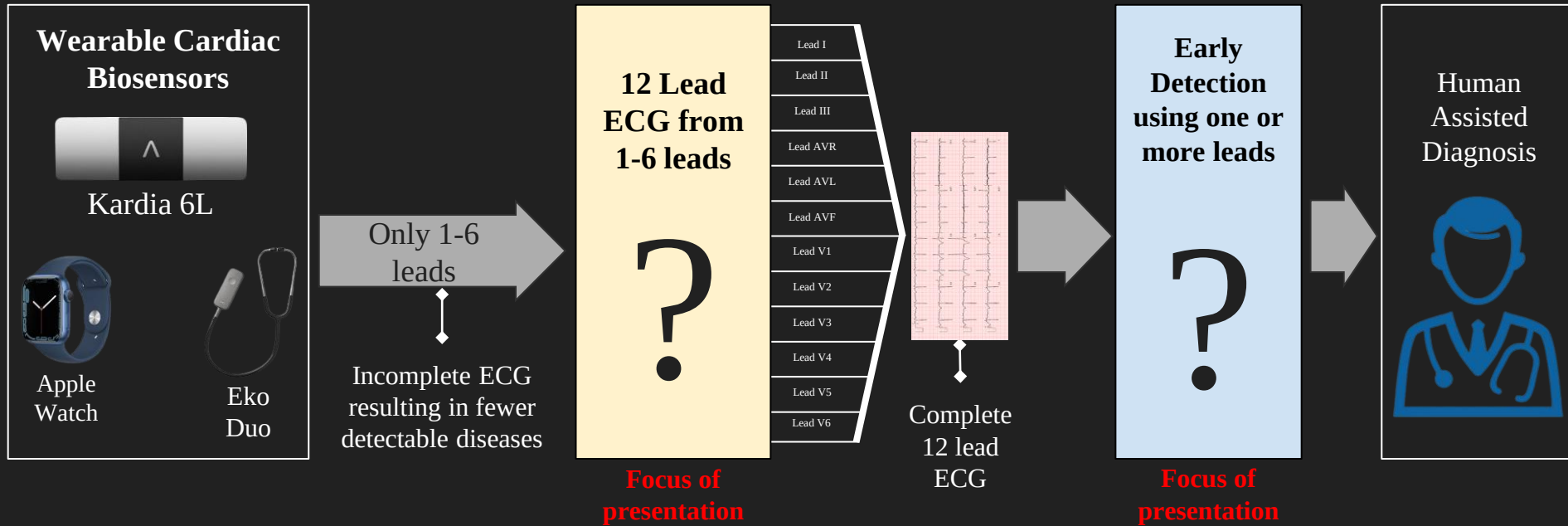


Early Detection and Synthesis of a 12 Lead ECG using Density-Based Clustering and Generative Adversarial Networks

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Problem Statement

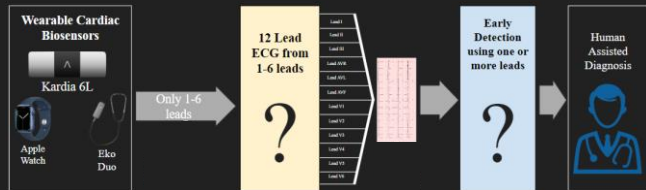
Create a AI based methodology for ECG synthesis and Early Detection, to detect a broad set of cardiovascular diseases before symptoms occur



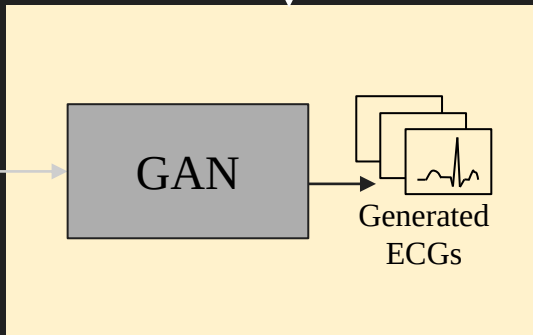
Concept Proposal

Training Data

- 12 Lead ECGs from PTB-XL dataset
- Biometrics
- Historical ECGs
- Personalized Heartbeat segmentation, feature extraction



Latent Vector
Recorded 3
Lead ECGs Eg.
Lead I, III, V6

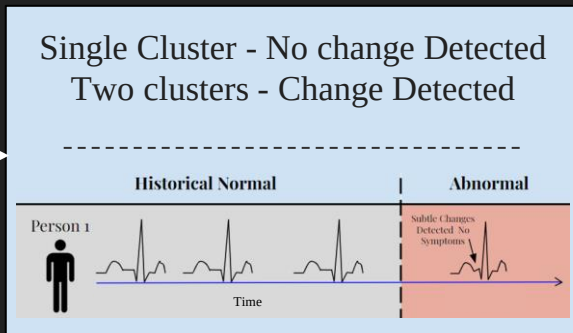


Personalized Generative Adversarial Network (GAN)

Train a Generative Adversarial Network to Synthesize a 12-Lead ECG from Leads I, III, V6:

- Create a 12 Lead ECG that is useful for accurate early detection and diagnosis

Synthesized
12 Lead ECG



Early Detection of CVD using Density-based Spatial Clustering For Applications With Noise (DBSCAN)

Use DBSCAN to detect changes in ECG

Detect changes by clustering historical and newly recorded ECG



Follow up visit with doctor if change is detected

Clustering Info from DBSCAN can be used as a starting point for complete diagnosis by doctor

**User Input,
Lead I, III, V2
into Trained
GAN**

The Right GAN Architecture For Synthesis

1

What Leads to Record?

	Lead	Eg. I, III, V6	How to record
1	I	Record	Recording of Einthoven lead I between the left arm wrist and the right index finger
2	II	Compute	Einthoven Law for (II = I + III)
3	III	Record	Recording of Einthoven lead III between the left lower abdominal region and the left index finger.
4	-aVR	Compute	Goldberger lead system (I+II) / 2
5	aVL	Compute	Goldberger lead system (I - III) / 2
6	aVF	Compute	Goldberger lead system (II + III) / 2
7	V1 (Wr)	Synthesize	Use GAN to synthesize
8	V2	Synthesize	Use GAN to synthesize
9	V3	Synthesize	Use GAN to synthesize
10	V4 (Wm)	Synthesize	Use GAN to synthesize
11	V5	Synthesize	Use GAN to synthesize
12	V6 (Wl)	Record	Recording of Wilson-like left (Wl) with the smartwatch at the fifth intercostal space in left midaxillary line.

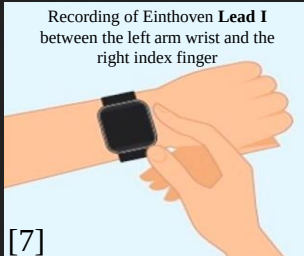
2

How are the Leads Recorded?

Recording of **Lead V6** with the smartwatch at the fifth intercostal space in left midaxillary line



Recording of Einthoven **Lead I** between the left arm wrist and the right index finger



4

Pick the Right Generator, Discriminator and Loss Functions

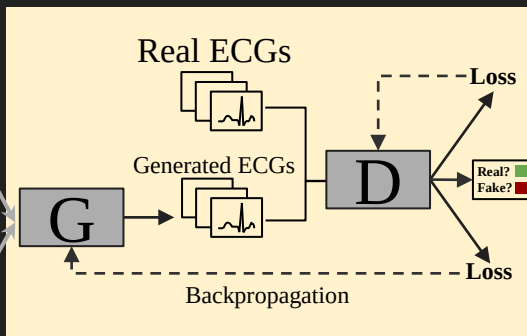
Generator

Discriminator

Loss Func

LSTM, GRU, BiGRU, CNN

BCE, Wasserstein, Least Squares



Which GAN Architecture to choose?

3

GAN Architectures Taxonomy

Extending

CGAN - Uses additional conditioning information, such as class labels or auxiliary variables, to generate data samples with specific characteristics

DCGAN - Utilizes deep convolutional neural networks as both the generator and discriminator

Info GAN - Learns interpretable and disentangled representations by maximizing the mutual information between selected latent variables and the generated samples

Scaling

WGAN - Uses the Wasserstein distance to provide more stable training and better convergence

Progressive GAN - Gradually increases the complexity and resolution of generated data during training, resulting in higher quality data

Style GAN - Learning and manipulating the latent space representation of the data, allowing for control over various aspects of the data's style

5

Pick the Right Hyperparameters and Parameters

Layers	Parameters	Hyperparameters
Convolutional Layer	Kernels	Kernel Size, Stride, # of Kernels
Pooling Layer	None	Pooling Method, Filter Size, Stride
Fully Connected Layer	Weights	Activation Function, # of Weights
Other		Model Arch, Optimizer, Learning Rate, Epochs

Early Detection Using Density Based Clustering

Historical Normal

Abnormal

Person 1

Subtle Changes
Detected
No
Symptoms

Time

Clustering Algorithm: Density-based Spatial Clustering for Applications with Noise (DBSCAN)

Explanation: Clustering algorithm that groups together data points based on their proximity and density

m_h = Normalized Mean PQRST Value of Historical ECGs

$$m_h = \sum_{i=1}^N (f_i)/N$$

f = Feature

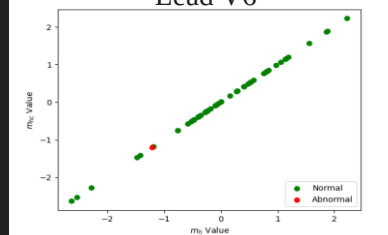
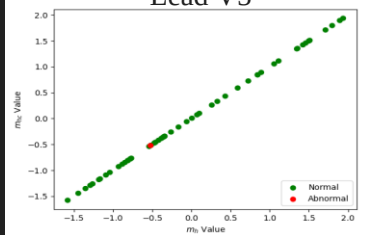
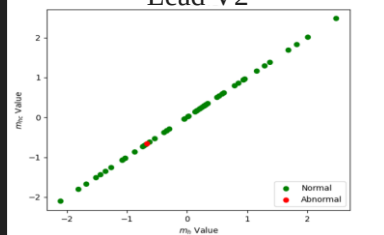
N= Total number of Features

Leads with Changes Detected

Lead V2

Lead V5

Lead V6



m_c = Normalized Mean PQRST Value of Current ECG

$$m_c = \sum_{i=1}^N (f_i)/N$$

f = Feature

N= Total number of Features

Leads with No Changes Detected

Lead I

Lead II

Lead III

Lead AVR

Lead AVL



m_{hc} = Normalized Mean PQRST Value of Historical and Current ECG

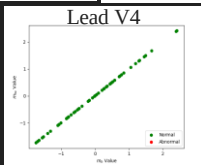
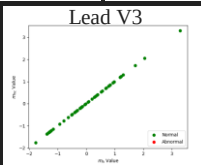
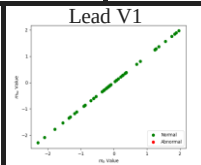
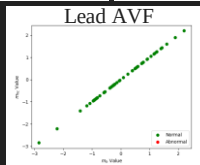
$$m_{hc} = (m_h + m_c)/2$$

Lead AVF

Lead V1

Lead V3

Lead V4



Input Data :

54 Historical (Normal) ECGs

1 current (LVH) ECG

Summary

Limited lead ECG will play a important role in the evolution of ECG recording techniques using wearable ECG recording devices. This will enable both extended duration, and multi scenario recordings which would otherwise not be possible

- Wearable Health Monitoring Devices are the most viable method for recording a ECG frequently, but can only record 1-6 leads
- A GAN with the right personalization is a very promising method to synthesize a 12 lead ECG from a limited lead set, and can be accurate enough for diagnostic purposes [2] [6] [8] [10]
- The use of DBSCAN clustering is an effective method for detecting changes in one or more leads

Future Work

- Personalization of GAN pipeline and training data will play a big role in achieving accurate synthesis of ECG
- Heartbeat segmentation, and Feature extraction for both 12 lead synthesis and Clustering may significantly improve accuracy enabling automated diagnosis for a broad set of cardiovascular diseases
- Standardized Figure of Merit (FoM) that is robust enough to drive confidence enabling clinical adoption

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